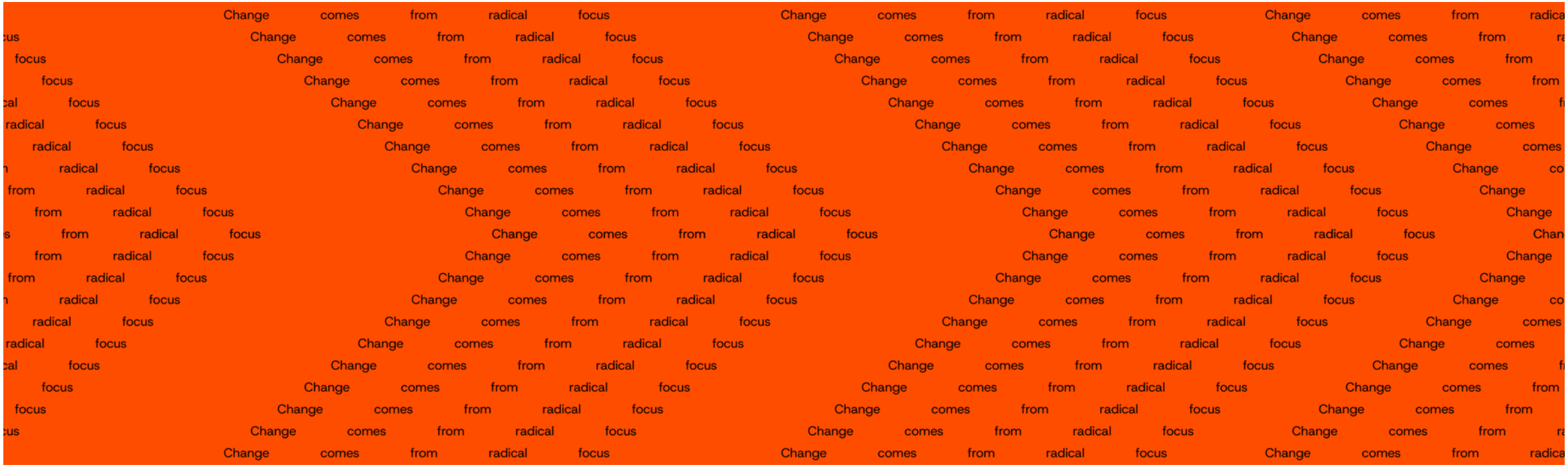
Velotic

ThingWorx Proficy Historian API Building Block

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Classification: Public

Purpose

Demonstrate Proficy Historian API connectivity to ThingWorx

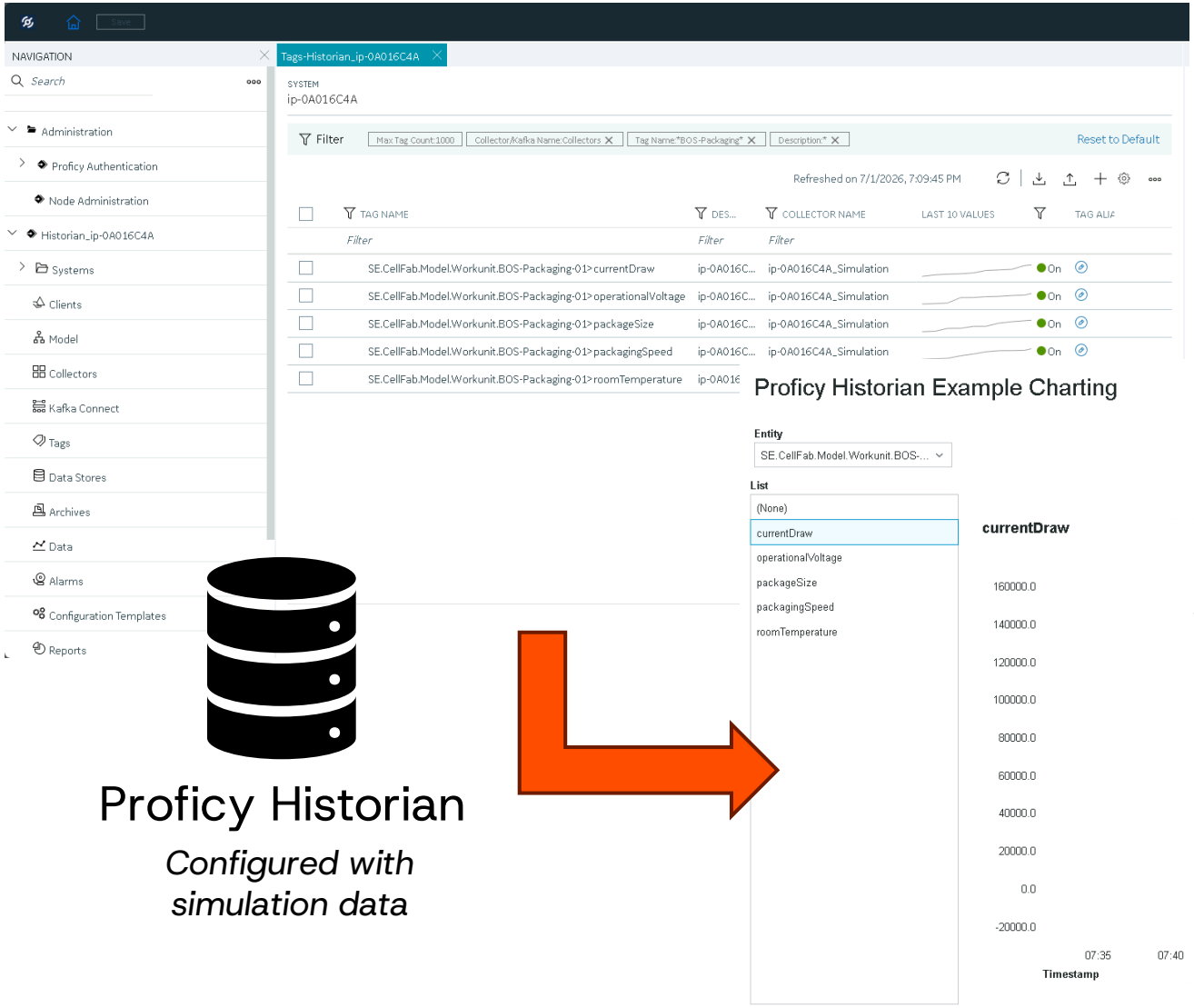
>Proficy Historian has a native REST API interface for querying time series data and information about the available tags.

>This Building Block is a proof of concept demonstration to authenticate and query time series data from Proficy Historian

>The associated Proficy Historian Charts Provider wraps the API for SCPA to provide assets with charting data.

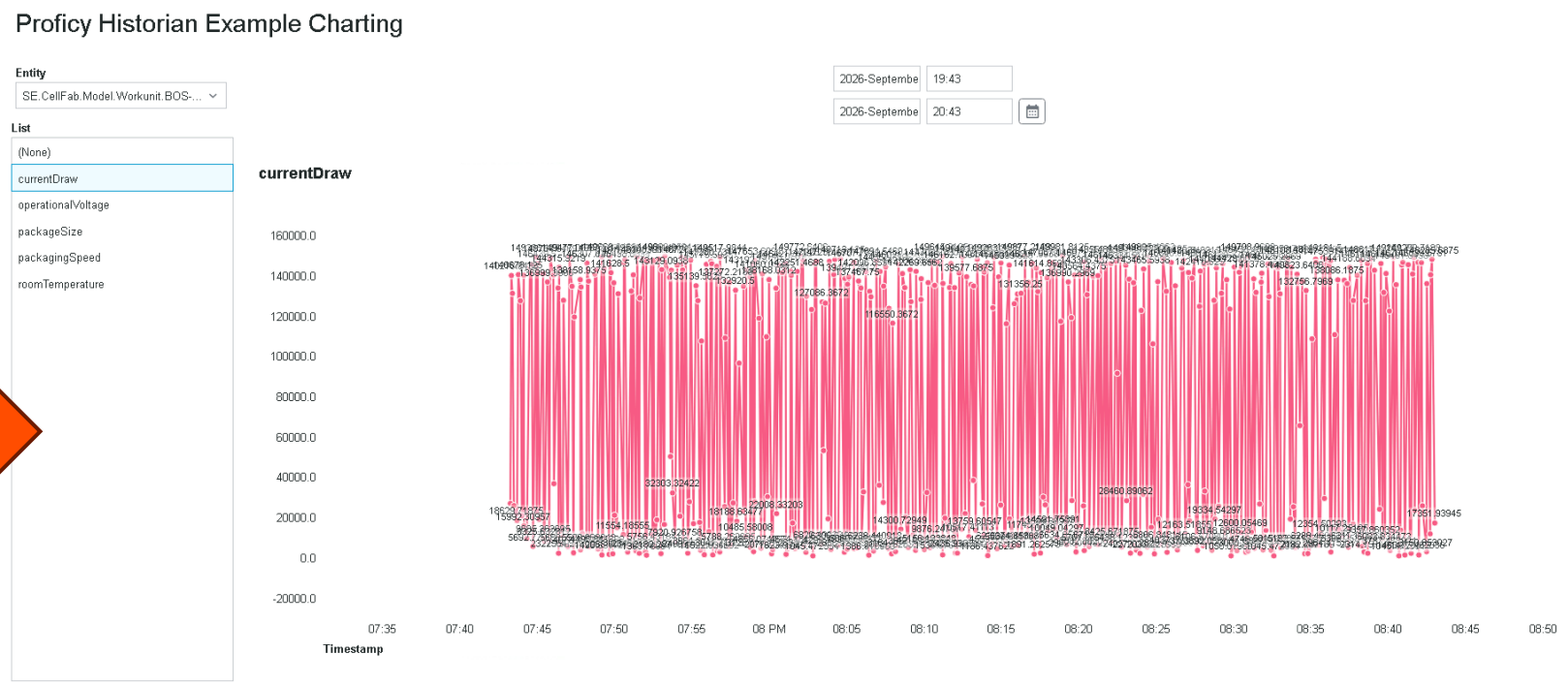
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Front-End Results



Proficy Historian

Configured with simulation data

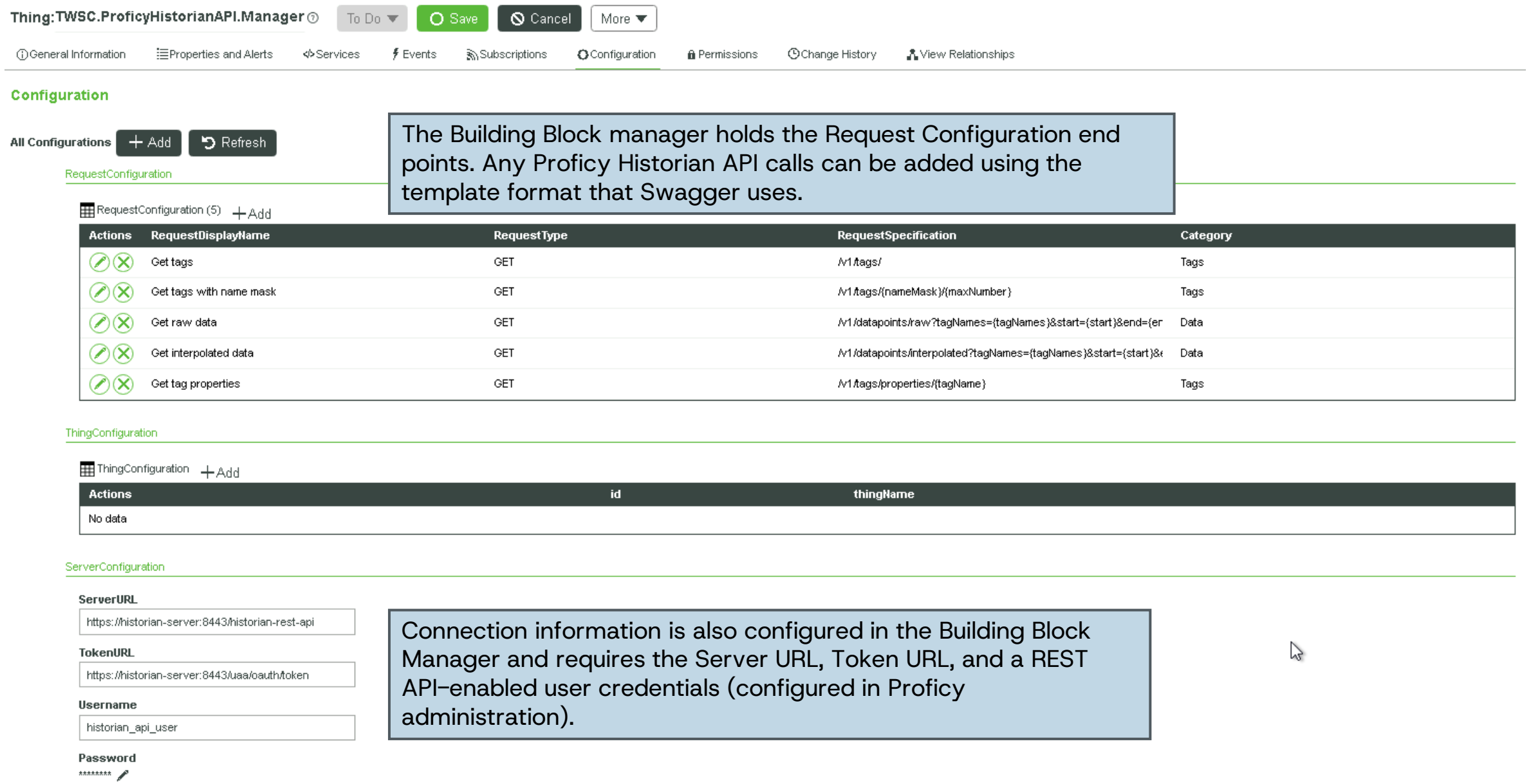


ThingWorx

Example Chart Mashup with Proficy Historian data

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Configuration



The Building Block manager holds the Request Configuration end points. Any Proficy Historian API calls can be added using the template format that Swagger uses.

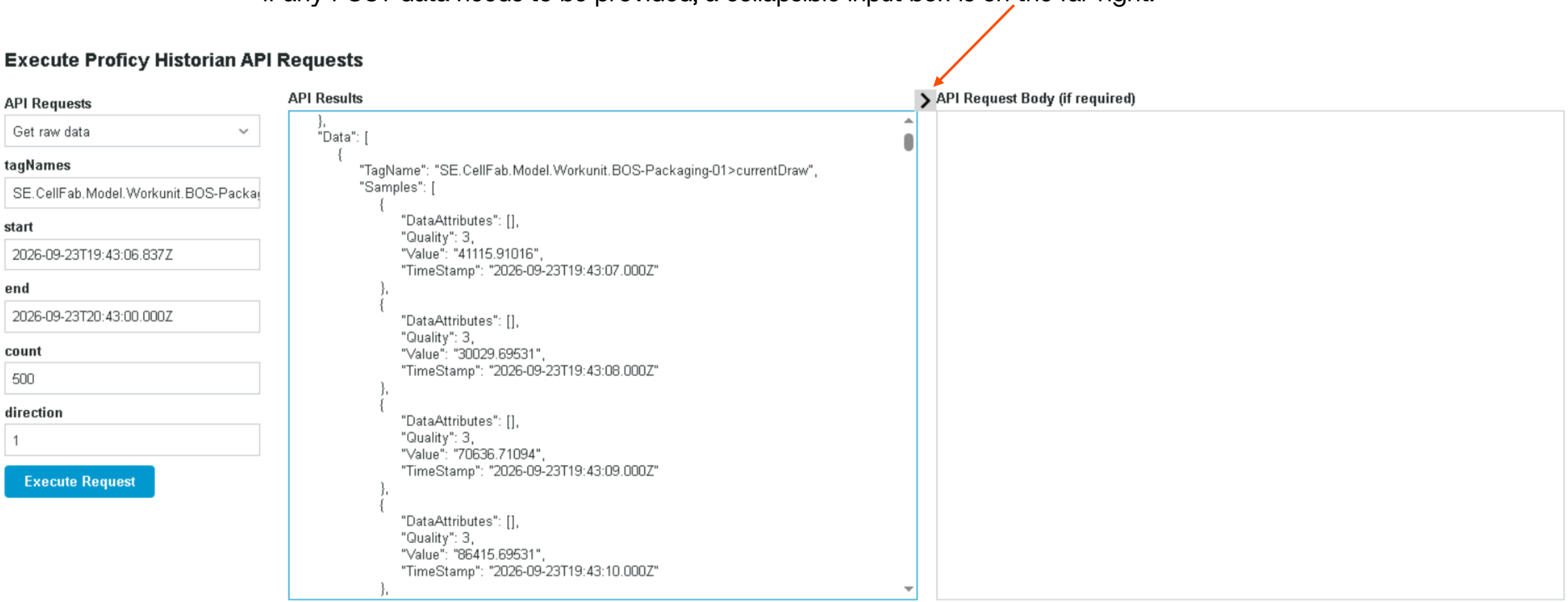
Connection information is also configured in the Building Block Manager and requires the Server URL, Token URL, and a REST API-enabled user credentials (configured in Proficy administration).

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Administration – Execute Requests

After an API Request is configured, the “Execute Requests” page can be used to execute the request and view the raw JSON response from the API. Input Parameter text fields are dynamically shown.

If any POST data needs to be provided, a collapsible input box is on the far right.

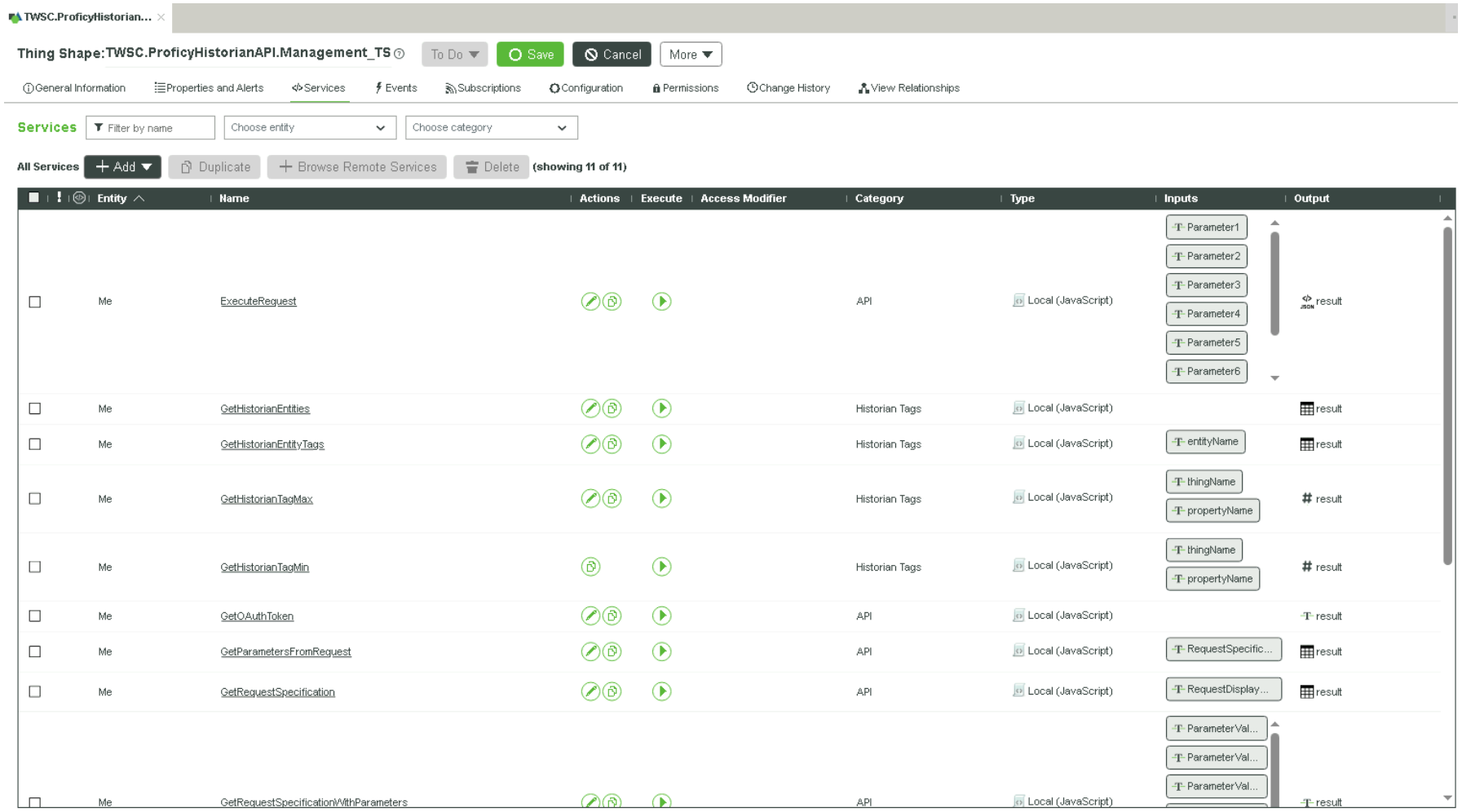


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Manager Thing Shape Services

Using the Configuration Tables as a reference for the requests, the services can flexibly call any Proficy Historian API and return JSON data.

The Manager automatically handles OAuth token fetching, caching, and renewal.



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Final Notes

This implementation was developed as a proof of concept and assumes:

> This Building Block assumed that Proficy Historian tag names will be in a scheme that fits with existing assets in a demo data set. No complex manipulation or mapping table of tag names was done.
> For the provided example services, the Historian tags must follow the `<EntityName>><PropertyName>` naming convention.
> The initial authentication token fetch can take several seconds.
  > If this were a production system, a scheduler might be created to refresh the token behind the scenes on a periodic basis.
> No more than six parameters are needed for request specifications
> String tag types were not tested due to limitations of our test environment.
> Scale testing was not performed.

Additional Historian deployments may require modifications to request definitions, authentication settings, or tag discovery logic.

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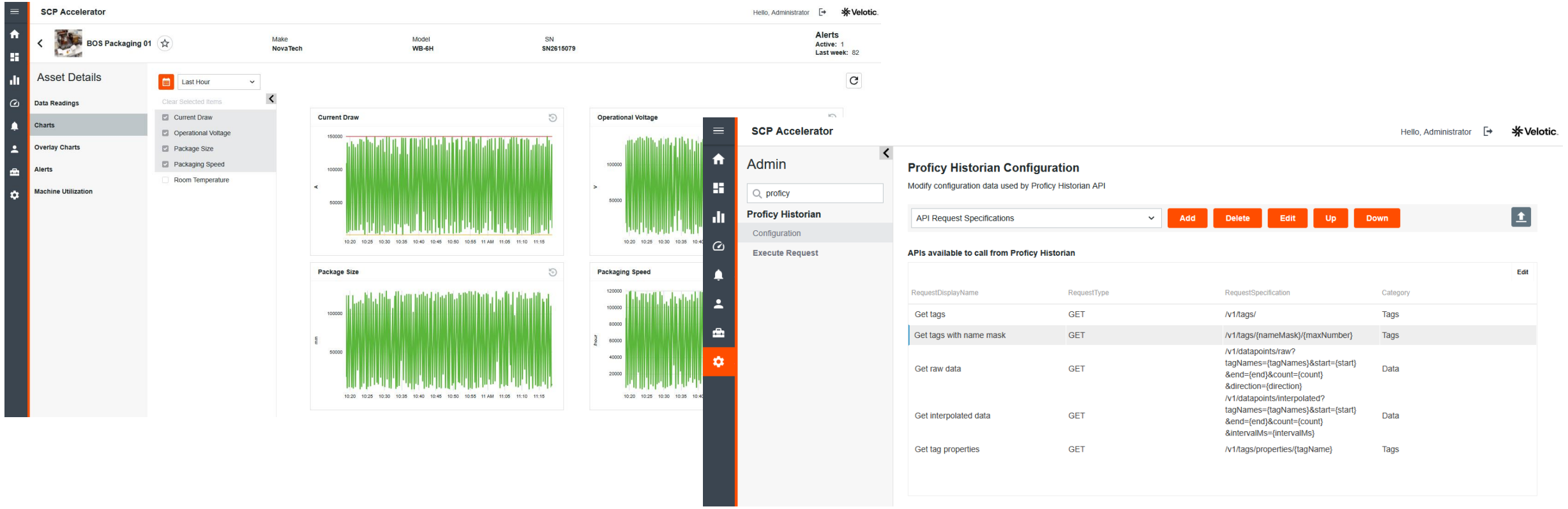
SCPA Charts Provider

An optional Charts Provider Building Block is also distributed with the Proficy Historian API Building Block.

> This Building Block provides a wrapper for SCPA to access the Proficy Historian data for any SCPA Asset that it is configured on. Additionally, it provides Admin Menu access to the Execute Requests mashup and a configuration page for the Request Configuration table.

> To enable, set the SCPA asset Thing's providerThing to use the TWSC.ProficyHistorianChartsProvider.Provider Thing.

> Tags in Historian must follow the `<EntityName>><PropertyName>` naming convention.



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